

# Barrier-Certified Seam Models for Compositional Robot Skills

One-page collaboration memo for Paper 119

**Core idea:** treat skill handoff as a local world/action model: predict whether composition will fail, diagnose why, choose repair/probe/abstain/transition, and update future planning.

**Problem.** Robot skills that work individually can fail when composed. A graph edge may look legal while the terminal state of the first skill falls outside the next skill’s attraction basin, crosses a high-energy barrier, or enters a contact mode where the next controller no longer descends. The failure is not only low-level control; it is a representation and prediction problem at the action seam.

**Method.** The proposed composer scores candidate handoff  $h$  by basin overlap, barrier height, descent continuity, repair cost, and calibrated seam risk:

$$S(i, j, h) = \alpha O_{\text{basin}} - \beta B_{\text{barrier}} + \gamma D_{\text{descent}} - \lambda C_{\text{repair}} - \kappa R_{\text{seam}}.$$

It then chooses among **accept**, **repair**, **probe**, **abstain**, or **fallback**. The contribution is a seam-level world/action model for behavior composition, not a new low-level controller or universal manipulation policy.

**Current evidence.**

| Hard-slice metric   | Proposed | Strongest non-oracle predecessor |
|---------------------|----------|----------------------------------|
| Success             | 0.80171  | 0.71711                          |
| Utility             | 0.88827  | 0.65310                          |
| Paired utility wins | 10/10    | —                                |

The frozen local suite covers 12 methods, 6 task families, 8 seam regimes, 5 deployment splits, and 10 paired seeds. The proposed method reduces seam failure, barrier violation, damage, risk calibration error, and realized seam breach while improving basin alignment and descent continuity. A diagnostic audit checks exported failure labels, seam decisions, and planner-edge updates over 230,400 local rows.

**Claim boundary.** The current claim is bounded: energy-seam certification improves skill composition when basin, barrier, descent, repair, and risk quantities are estimable. The paper should not claim deployment-ready robotics until it has real-robot or accepted high-fidelity validation with shared skill libraries, paired resets, raw logs, videos, and independent baselines.

**Why this is a natural fit for behavior composition.** CoStream asks how simple behaviors can compose into complex manipulation. Paper 119 asks when a composed behavior handoff should be trusted, repaired, probed, or avoided. A clean collaboration question is whether a seam model improves robustness for a CoStream-style stack under perturbations, precision assembly, object transfer, and contact-mode transitions.

**Proposed validation.** Use the same primitive skills across baselines; log terminal samples, basin estimates, barrier scores, descent diagnostics, predicted risk, accepted/rejected seams, outcomes, and videos. Compare against option graphs, TAMP screens, diffusion stitching, CEM, residual repair, energy heuristics, the previous v4.1 method, and an oracle post hoc upper bound.